Pocket TTS Learns Japanese: Training Data and Distillation for a Drop-in 100M-Parameter Model

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https://github.com/takuma104/pocket-tts-ja

Abstract

Pocket TTS is a 100M-parameter text-to-speech (TTS) model from Kyutai that runs faster than real time on a laptop CPU: a causal transformer predicts the continuous latents of the Mimi codec, and a small flow head samples each 80 ms frame in one step. We adapt it to Japanese under a strict constraint: the result must load in the stock `pocket-tts` package through a config file, read plain kanji–kana text without a pronunciation front end, and keep zero-shot voice cloning. Most of the work went into the training data. We filter web speech (Emilia, YODAS3) with ASR re-checks, restore punctuation, match loudness with per-speaker gain copies, and cover rare kanji readings with 0.4–0.8 kh of synthetic speech whose readings an ASR model confirmed. Accuracy follows how often a reading occurs in training, and new synthetic data, not more steps, broke every plateau. The stock runtime generates a few frames after the end-of-speech decision that upstream training never supervises; supervising them with digital silence removed fillers that followed about 10% of the sentences. Guidance is distilled into the weights, and a 6-layer student reads better than its 24-layer teacher. Finally, only the codec's decoder is fine-tuned, so latents and voice prompts stay compatible. On the 13,536 sentences of JKYB-Parakeet, the released 6-layer models read 94.75% (trained with Emilia, CC BY-NC) and 94.77% (CC BY data only) of the target words correctly.

1 Introduction

Pocket TTS [1] is a small TTS model released by Kyutai in 2026 and built on continuous audio language models (CALM) [2]. It has about 100M parameters, streams with about 200 ms of latency, runs about 6× faster than real time on two laptop CPU cores, and clones a voice from a few seconds of audio. Kyutai publishes its training code, including a recipe for new languages. The official models cover English and six European languages; none covers Japanese.

Japanese is hard for such a model. Text has no spaces, and most kanji have several readings that depend on the word and the context: 生 is *sei* in 生活, *nama* in 生ビール and *i* in 生きる. Pocket TTS has no text encoder and no dictionary: a lookup table embeds the tokenizer's pieces, and the transformer must learn every reading from paired speech, in which rare readings occur only a few times.

Our goals were: (i) the model loads in the stock package (3.3.0) with `--config` and no code changes; (ii) it reads plain Japanese text; (iii) a 6-layer model runs on CPU; (iv) besides a line trained with the non-commercial Emilia corpus, a second line uses CC BY data only, so that its weights can be used commercially. We describe the training data (§3), the training recipe and the supervision of the utterance end (§4), distillation (§5), the decoder (§6) and the released models (§7), which are on Hugging Face[1].

2 Pocket TTS

FlowLM. A causal transformer (6 or 24 layers, width 1024, 16 heads) reads the conditioning and then the target latents frame by frame. Its output conditions a flow head, an MLP with adaptive layer norm (6 blocks, width 512) [3] trained with Lagrangian self-distillation (LSD) [4], a consistency-style objective. Inference takes one step, $x_1 = x_0 + v(0, 1, x_0)$ with $x_0 \sim \mathcal{N}(0, \tau)$ and temperature $\tau = 0.3$. A linear head on the transformer output predicts the end of speech (EOS). The sequence is a learned token, the voice prompt (projected Mimi latents of the reference audio), the text (one embedding per piece), a BOS latent and the target. Training drops the text and the voice 20% of the time each, for classifier-free guidance (CFG) [5] on the transformer output.

Mimi, the codec of Moshi [6], encodes 24 kHz audio into 32-dimensional continuous latents at 12.5 Hz with a convolutional SEANet encoder and decoder (strides 6, 5, 4) and small transformers. It is frozen and the same in every language release (Table 1).

*Claude (Opus 5.5), working in Claude Code, planned and implemented the data pipeline and the experiments, ran them, analyzed the results and wrote this report; a second Claude Code session did the decoder fine-tuning of §6. Takuma Mori set the goals (a Japanese model that the stock `pocket-tts` package loads unchanged, and a second line trained on CC BY data only), chose the data sources and their licenses, provided the machines, did the listening tests (which found the extra speech after short sentences and chose the decoders) and decided the releases.

[1] https://huggingface.co/takuma104/pocket-tts-ja (CC BY-NC 4.0) and https://huggingface.co/takuma104/pocket-tts-ja-cc (CC BY 4.0).

Table 1: Parts of Pocket TTS (6- / 24-layer models) and what this work trains (*frozen during distillation).

Part	Params	This work
Transformer (width 1024)	75.5M / 302.1M	trained
Text table (4,000 pieces)	4.1M	new
Flow head, EOS head	9.8M	trained*
Mimi encoder side	9.8M	frozen
Mimi decoder side	10.3M	fine-tuned (§6)
Total	109.5M / 336.1M	

What the stock runtime fixes. The config sets the weights, the sizes, the SentencePiece tokenizer, the temperature, text replacements and the number of frames after EOS. The rest is fixed: there is no CFG; text is cut into chunks of at most 50 tokens at sentence, then clause punctuation; generation stops at tokens/3 + 2 s; the tokenizer does no normalization. After EOS the runtime generates 3 more frames for inputs of up to four space-separated words (else 1), plus 2, so a Japanese sentence, one "word", gets 5 frames (400 ms). These rules explain a small vocabulary, ASCII punctuation in training, distilled CFG, and our care for short chunks and the frames after EOS.

3 Training data

3.1 Natural speech

Emilia [7, 8] has 1,716 h (Emilia, CC BY-NC) and 1,120 h (Emilia-YODAS, CC BY) of Japanese 3–30 s clips. Three Emilia shards are 95% byte-identical to Emilia-YODAS, and 21% of the rest are duplicates (re-uploaded listening drills, channel outros). A speaker label is one diarized speaker in one video (ECAPA [9] similarity 0.69–0.73 within a label, 0.15 across), so the voice prompt, which must be another utterance of the target speaker, is taken from the same label. After de-duplication and filters (1.5–15 characters/s, DNS-MOS [10] ≥3.0, no benchmark sentences, ≥2 utterances per speaker, speaker consistency), 1,709 h remain.

YODAS3 [11, 12] (CC BY 3.0) has 17,404 h of full-band Japanese YouTube audio with no segments, speakers or quality scores and mostly automatic captions without punctuation. We cut 3–20 s utterances at pauses (Silero VAD [13]), transcribe them with Whisper [14] and keep an utterance when the readings of caption and transcript agree (kana CER ≤0.10 without fillers); the transcript is the label. Speakers are per-video ECAPA clusters; a cluster that matches clusters in ≥20 other videos is a mass-produced TTS voice (in a pilot, all were Wikipedia read by TTS) and is dropped. Of 8,311 h of segments, 601 h remain. 72% of the labels had no punctuation (Emilia 2–7%), which teaches pauses without commas; transcribing each clip again with a punctuated prompt and copying only the punctuation leaves 1.3% without.

3.2 What listening and ASR found

Speech after short sentences. Listening found junk after short sentences (…ハチガツデス、ゲツ): 26–44% of the JKYB sentences under 15 kana ended with two or more extra kana, none above 30. The shortest Emilia clip is 3.04 s, so short transcripts sit in long clips (2.2 against 5.6 characters/s), and in 40% of those we heard, the speaker repeated the sentence or kept talking. We decode the exact training latents, transcribe them with kana-whisper [15] and drop a clip if the ASR inserts $\geq \max(3, 0.2n)$ kana after removing fillers (n: reference kana), the CER exceeds 0.3 or the ASR loops. Emilia keeps 1,531 h, and with short Tatoeba sentences (S4 below), extra speech after sentences under 15 kana fell from 25.8% to 3.0%. On YODAS3 the check dropped half of the clips under 15 kana (559 h remain).

Table 2: Synthetic sets of the main line. Kept: passed the ASR check of every target and of the sentence.

Set	Texts	Kept	Hours
S3	S1 (targeted, Wikipedia, general)	179k (60%)	341
S4	Tatoeba, 2–20 characters	21k (54%)	19
S5	S3 rejects, target in katakana	43k (55%)	84
S6a/b	Wikipedia, readings to 300 sentences	108k (72%)	215
S7	LLM sentences, rare readings*	13k (82%)	15
S9	2–8 tokens, explicit duration	31k (69%)	11

*Written by Claude Haiku; homographs checked by Claude Sonnet.

Loudness. Quiet prompts gave quiet outputs: training audio sits at −24 to −14 LUFS, JVS prompts at −24 to −31, and the runtime does not normalize prompts. We re-encode 20% of the Emilia clips with a per-speaker gain from $[-15, +3]$ dB, as separate speakers so that prompt and target share it (227 h); the loudness gap to the prompt went from −2.2 to −0.9 LU.

3.3 Reading-rich synthetic speech

Why. With Emilia only, the teacher stalled at 70–72% on JKYB-mini. Accuracy follows the exposure of a kanji–reading pair, the number of training sentences that contain it: 32% for 1–9, 54% for 10–29, 83% for 30–99, 89% for 100–299 and 95–96% above. Writing the target in katakana raised accuracy from 88.1% to 96.2%: the model can say the readings but lacks the mapping from kanji.

Teacher and texts. We synthesize with Irodori-TTS-v4.1-Small-Yomi [16], whose kanji readings we had improved by text-only self-distillation (JKYB 95.7%, MIT license), in 3,000 voices from Emilia. The first 300k sentences (S1) are the reading-targeted sentences of that work (KANJIDIC2 [17], Aozora Bunko ruby, LLM sentences), Wikipedia sentences with words rare in Emilia, and general sentences. No word was chosen with the benchmark's keys, and sentences that overlap JKYB, JVS, JSUT or the Aozora evaluation works were removed.

Quality control and its bias. A clip is kept only if kana-whisper reads every target exactly and the sentence CER is ≤0.08. This removes exactly the hard cases: of 120k rejected S3 sentences, 84k failed because Irodori itself misread the target. Set S5 re-synthesizes them with the target written in katakana, keeping the kanji transcript. Later sets top up readings that stayed rare (Table 2).

Table 3: Training mixes of the released teachers (hours). NC: CC BY-NC 4.0; BY: CC BY (*by the reference voices).

Source	Main	CC BY	License
Emilia (CC BY: Emilia-YODAS only)	1,531	305	NC / BY
gain copies	227	48	
YODAS3	–	556	BY 3.0
Synthetic (Irodori-TTS)	828	425	NC / BY*
Total	2,592	1,334	

Effect. Each set moved the plateau (Fig. 1). 10% of S3 took JKYB-mini from 72.3% to 77.2% within 7k steps (JSUT CER 6.95→5.36%). S7 raised the readings it topped up with ≥10 sentences from 53% to 80%; readings it did not touch stayed at 50%. S6a and S5 followed (91.0%, 91.8%). Speaker similarity did not change with 22–31% synthetic data. For the CC BY line, the 309k sentences that had passed in Emilia (NC) voices were synthesized again with the same text and seed in Emilia-YODAS voices (2,998 voices, 553 h).

3.4 Text, tokenizer and mixes

Text is normalized (NFKC, 。 →., 、 →,, no brackets, a space after each sentence end, as the runtime joins chunks with spaces), and 124 replacement rules in the config do the same at inference. The tokenizer is a 4,000-piece SentencePiece unigram model [18] trained on the transcripts. Against 8,000 pieces it read better after 15k steps (65.3 vs 63.4%, McNemar $p = 0.031$), and its 50-token chunks fit the length limit more often. The runtime may generate a sentence's last clause alone, but only 0.04% of the training utterances have five tokens or fewer, and Irodori stretches very short texts to 2–3 s; S9 synthesizes 2–8-token texts with an explicit duration (0.13 s per mora). Table 3 gives the final mixes. A script checks every row of the CC BY mixes against the source manifests: none comes from the NC part of Emilia, directly or as a synthetic voice.

4 Training the teacher

Recipe. We use the upstream training code unchanged and replace only the data loader. The English 24-layer model is the start, with a new text table. AdamW (weight decay 0.1), learning rate $2 \cdot 10^{-4}$, batch 64, targets up to 30 s. The voice prompt is another utterance of the same speaker: a 3–10 s prefix (50%), a whole utterance (35%) or two (15%). One RTX 5090 runs 2.5 steps/s; the main teacher's 700k steps took 76 hours.

Data, not steps. The main teacher plateaued three times (70–72%, 83.5–85%, 87.5–88%), and each plateau ended only when a new synthetic set entered (Fig. 1). The validation loss did not show the gains: it was the same at 93k and 100k steps while JKYB-mini rose by 5 pt. A final cosine decay (600k→660k) added 0.6 pt on JKYB-mini and 4.6 pt on Aozora (JSUT CER 2.86→2.40%). The CC BY teacher, with synthetic data from the start, reached 90% at 35k steps.

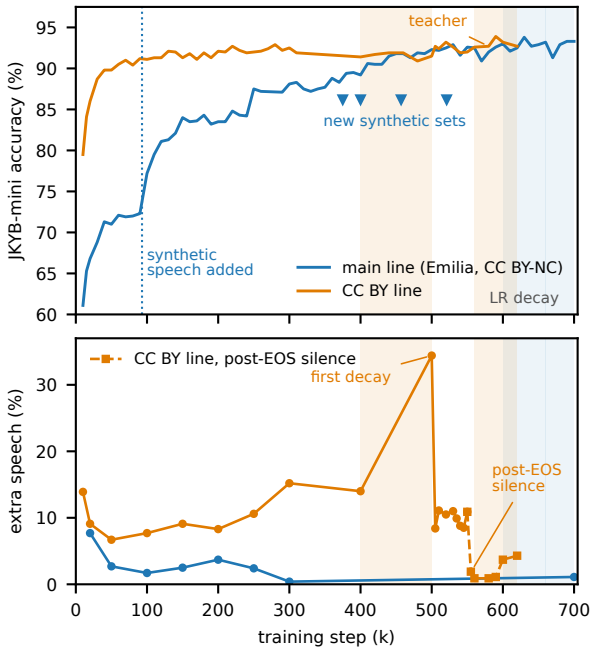


Figure 1: Teachers during training (CFG 2.0). Top: reading accuracy rises when new synthetic data arrives, not with more steps. Bottom: sentences with extra speech after the text. In the CC BY line, data cleaning and decays did not remove it; supervising the frames after EOS did. Shaded: decays.

4.1 How an utterance ends

Short chunks. On 172 short texts the main teacher at 660k added speech after 58 and did not stop on 22 (CER 28%). A short stage with S9 at 1% of the hours (40k steps, learning rate raised and decayed again) fixed both within 5k steps (1 and 1, CER 4.8%).

Frames after EOS. The CC BY teacher added fillers ("うん", "えー") after 7–15% of the JKYB-mini sentences, 3–5× the main line; its first decay raised this to 34%. Data fixes each stopped near 7–11%: trimming YODAS3 clip ends, re-checking YODAS3 by ASR, dropping fillers, even removing YODAS3 (7.1%, with speaker similarity 0.545→0.500). Generating no frames after EOS gave 3.3% instead of 8.8%: most fillers were in the 5 frames the runtime adds. Upstream training masks the loss after the last speech frame, so these frames are never seen. We append six frames of digital silence (the Mimi latents of zeros) to every complete target and apply the flow loss to them; the EOS target stays where speech ends. Within 10k steps extra speech fell from 10.9% to 0.9% (main line 1.1%), sentence CER from 3.6–3.9% to 1.9%, short-text CER from 11.8% to 5.2%, with no change in EOS loss or speed. Late in the next decay fillers returned, now before EOS (4.3% at 620k), so the teacher is the 580k checkpoint.

5 Distillation

Guidance. Without CFG the teacher read 1.2 pt worse, failed to stop more often and was 3.9 LU quieter; scale 2.0 read best (93.0% vs 90.5% at 2.5). Since the runtime

has no CFG, a copy of the 24-layer teacher learns the frozen teacher's guided state $z_\varnothing + 2(z_c - z_\varnothing)$ (MSE on the transformer output, heads frozen, 50k steps). It reads slightly better than the guided teacher and is released as `japanese_24l`.

Depth. The 6-layer student follows upstream's depth distillation: it starts from the teacher's layers 0, 1, 2, 21, 22 and 23 (upstream found the outer layers the best seed) and learns the guided state (main line: 42k + 120k steps; CC BY: 160k). On all of JKYB the students *beat* their teachers: 94.15 / 94.73% against 92.97 / 93.68% (main) and 93.93 / 93.96% against 92.82 / 93.82% (CC BY; prompts in rotation / jvs001, original decoder), as upstream's English student does (WER 0.76 vs 0.82%). They fail to stop more often (43 of 13,536 sentences against 6). In bfloat16 the 6-layer model runs 4.4× faster than real time on one CPU thread (24 layers: 1.4×).

Frames after EOS at release. A level check of the clip ends showed that half of the 24-layer CC BY model's short end sounds come after EOS (10.1% of clips; 2.5% cut while sounding). With 3 frames they drop to 7.9% with 0.7% cut (0 frames: 5.9% but 16% cut), so its config sets 3. The 6-layer model's sounds come before EOS (9.4–9.8% at any setting) and listening found them unobjectionable, so it keeps the default.

6 Fine-tuning the decoder

Why. Through Mimi, the kana CER of all JVS [27] read speech rises from 3.16% to 5.45%, mostly by consonants that change place (キ → チ, ガ → ダ). A consonant classifier gets 94.1% from the latents but 92.0% from the decoded audio: the decoder loses the information. We train only the decode side (10.3M parameters), so the encoder, latents, language model and voice prompts stay as they are, and the runtime loads the decoder from the same weight file.

Main line. Losses: multi-scale mel, MPD [19] and MS-STFT [20] discriminators, and a loss on the batch-averaged spectrum against tones at the stride frequencies (multiples of 200 Hz and 1.2 kHz); data: 182 h of synthetic speech and 60 h of high-bitrate Emilia. Every metric improved, yet listening rejected two decoders, for metallic consonants and a pitch-locked high band: above 4 kHz they made noise where recordings have harmonics. Holding that band at the original decoder's output for the last 10k steps (*keep4k*) fixed it by ear and kept the gains (Table 4).

The high band. A research branch trained the band above 4 kHz towards studio recordings [21, 22]. Only UTMOSv2 [23] (40 Hz high-pass on every file) agreed with all listening verdicts (UTMOS22 [24]: rank correlation −0.86; ViSQOL [25] ranked a rejected decoder first), and even it missed a mechanical sound. Harmonic excitation from an F0 head sounded worse and needs new inference code; wider decoders changed nothing. The predicted MOS of the training data set the order of the decoders, and averaging runs of different recipes [26] gave the best one, which uses NC and share-alike data.

CC BY decoder. It may use only the original Mimi, YODAS3 and the synthetic speech in CC BY voices. YODAS3 clips are cut again at 24 kHz from videos with audio up to 15 kHz and high DNSMOS (90k clips, 234 h). Ranking clips by UTMOSv2 helped, by harmonic strength not. The base trains 70k steps from the original Mimi on 164 h of YODAS3 and 116 h of synthetic speech, discriminators included; four 10k-step runs then train the high band on the top UTMOSv2 clips without band limits or clipping (7–14 h) and are averaged. Two videos that listening revealed as synthetic voices were left out. The result matches the best NC decoder in reconstruction and slightly exceeds it in TTS (UTMOSv2 +0.35–0.42 over the original Mimi, speaker similarity 0.55→0.65); in listening, the CC BY decoders could not be told apart from it.

Table 4: Decoders on JVS reconstruction (10 speakers × 100 sentences). UTMOSv2 relative to keep4k.

Decoder	CER	Sim.	UTMOSv2	Data
Recording	3.34	–	+0.400	
Original Mimi	5.44	0.698	−0.036	
keep4k (main release)	4.70	0.779	0	NC
Best average (research)	4.74	0.784	+0.397	NC, SA
CC BY (`M2CH2_soup4`)	4.69	0.805	+0.383	CC BY

Table 5: Released models (GPU evaluation; L: layers; teachers with CFG 2.0). JKYB: all sentences, prompts in rotation / jvs001. CER in %. Short: did not stop / extra speech, of 172. Red: provisional.

Model	JKYB	mini	Aoz.	JSUT	JVS / sim.	Short
Main line (CC BY-NC), decoder keep4k						
6L	94.75 / 94.84	95.1	71.3	2.28	2.32 / .577	0 / 1
24L	94.32 / 94.49	94.5	71.2	2.28	2.13 / .586	0 / 1
CC BY line, CC BY decoder						
6L	94.77 / TBD	94.9	70.7	2.14	2.18 / .651	0 / TBD
24L	TBD	TBD	TBD	TBD	TBD	TBD
Original Mimi decoder						
Main 6L	94.15 / 94.73	93.8	70.0	2.39	2.58 / .536	0 / 1
CC BY 6L	93.93 / 93.96	94.3	69.0	2.40	2.47 / .560	0 / 1
CC BY 24L	93.34 / 94.08	93.5	68.5	2.48	2.67 / .547	0 / 0
Main teacher	92.97 / 93.68	93.3	69.4	2.66	3.05 / –	1 / 2
CC BY teacher	92.82 / 93.82	92.7	67.9	2.58	2.45 / .546	0 / –

7 Released models

Table 5 compares the releases. Generated audio is transcribed with kana-whisper: JKYB-Parakeet [28, 29] (13,536 sentences, each with a tagged target; prompts from four JVS speakers in rotation, or jvs001), its 1,500-sentence subset JKYB-mini, 1,000 Aozora ruby targets from works not used for synthesis, 1,000 JSUT [30] sentences, zero-shot cloning of 10 JVS speakers (kana CER and ECAPA similarity to the speaker's own recording of the sentence) and 172 short texts. JVS and JSUT are never trained on. The stock package on CPU, loading the released files, matches within noise (main line `japanese`: JKYB-mini 95.3%, JSUT 2.15%, JVS 1.70% / 0.575).

With its own decoder the CC BY line, trained on half the data, reaches the main line's accuracy. Both 6-layer models misread about 5% of the JKYB sentences, mostly rare readings and jukujikun.

8 Discussion

What mattered. First, *coverage*: the model learns a reading only when it hears it often enough, and synthetic speech from a TTS model that reads well, checked by ASR, is the cheapest way to provide it, provided the check keeps the hard readings. Second, *train what the runtime does*: the frames after EOS, short chunks and the missing CFG each had to be supervised or distilled. Third, *metrics are not enough for the decoder*: every rejected decoder improved all of them.

Limitations. Scores are ASR-based with one seed, and JKYB-mini moves by about ±0.5–1 pt between evaluations. We ran no controlled ablation of the synthetic share, and the synthetic speech carries Irodori's readings and style. The students fail to stop slightly more often than the teachers, and 8–10% of the CC BY clips end in a short, quiet sound.

9 Conclusion

A drop-in Japanese Pocket TTS needs no change to the runtime. Most of the gain came from the data: web speech re-checked by ASR, matched loudness, and synthetic speech that covers rare readings. Supervising the frames generated after EOS removed trailing fillers, distillation gave a 6-layer model that reads better than its teacher, and a decoder fine-tuned on CC BY audio brought the CC BY line level with the line trained with Emilia.

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